## Type MS · Logical and arithmetic binary shifts

### Answer 1 of 5 · Question 2 part c.ii · 1 mark

2023 JUN · Paper 01 · extract 1 of 5 · find part c.ii in this mark scheme extract

<table>
<tr>
<td>2(a)</td>
<td>Award one mark for each correct cell:
<table>
<tr><th>Denary</th><th>Hexadecimal</th></tr>
<tr><td>8</td><td>8 (1)</td></tr>
<tr><td>12</td><td>C (1)</td></tr>
</table>
</td>
<td></td>
<td>2</td>
</tr>
</table>

## Type MS · Logical and arithmetic binary shifts

### Answer 1 of 5 · Question 2 part c.ii · 1 mark

2023 JUN · Paper 01 · extract 2 of 5 · find part c.ii in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(b) | **The only correct answer is D**<br><br>***A*** *is not correct because* $6^2$ *stores 36 patterns, not 64 patterns*<br><br>***B*** *is not correct because* **6 × 2** stores 12 patterns, not 64 patterns<br><br>***C*** *is not correct because* $6^2$ **– 1** stores 35 patterns, not 64 patterns | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(c)(i) | Award **one** mark for:<br><br>• 0001 0111 (1) | Ignore spaces | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(c)(ii) | Award **one** mark for:<br><br>• 1111 1000 (1) | Ignore spaces | 1 |

## Type MS · Logical and arithmetic binary shifts

### Answer 1 of 5 · Question 2 part c.ii · 1 mark

2023 JUN · Paper 01 · extract 3 of 5 · find part c.ii in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(d)(i) | Award **one** mark for each correct nibble:<br>• 0011 1100 | Ignore spaces | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(d)(ii) | Award **one** mark for each of:<br>• Negative sign (1)<br>• 22 (1)<br><br>Example:<br>• -22 | Note 22 is calculated from 16 + 4 + 2 | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(d)(iii) | Award **one** mark for each correct nibble:<br>• 0001 1011 | Ignore spaces | 2 |

## Type MS · Logical and arithmetic binary shifts

### Answer 1 of 5 · Question 2 part c.ii · 1 mark

2023 JUN · Paper 01 · extract 4 of 5 · find part c.ii in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(e) | Award **one** mark for each of:<br><br>1. 13 x 1000 // 13000 in the numerator (1)<br>2. 1024 in the denominator (1)<br><br>Examples:<br>$\frac{13 \times 1000}{1024}$<br><br>$\frac{13000}{1024}$<br><br>$\frac{13 \times 500}{512}$<br><br>$\frac{13 \times 250}{256}$<br><br>$\frac{13 \times 125}{128}$<br><br>$\frac{13 \times 10^3}{2^{10}}$ | Award reduced expressions, if correct<br><br>Allow all equivalent exponents<br><br>Note 1000/1024 = 1.024, so could have 13 * 1.024 | 2 |

## Type MS · Logical and arithmetic binary shifts

### Answer 1 of 5 · Question 2 part c.ii · 1 mark

2023 JUN · Paper 01 · extract 5 of 5 · find part c.ii in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|

Type MS · Logical and arithmetic binary shifts

Answer 2 of 5 · Question 1 part e · 1 mark

2021 JUN · Paper 01 · extract 1 of 2 · find part e in this mark scheme extract

| Number | | | |
|---|---|---|---|
| 1(a) | **The only correct answer is B**<br><br>***A*** *is not correct because computers only understand binary*<br><br>***C*** *is not correct because Hex values will be held as binary so take up the same space*<br><br>***D*** *is not correct because Hex values are held as binary and so take the same time to execute* | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(b) | Award **one** mark for each of:<br>• 5<br>• A | Must be in the correct order | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(c) | **The only correct answer is B**<br><br>***A*** *is not correct because this is the number of binary patterns that can be represented by 7 bits*<br><br>***C*** *is not correct because this is the number of binary patterns that can be represented by 9 bits*<br><br>***D*** *is not correct because this is the number of binary patterns that can be represented by 10 bits* | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(d) | Award **one** mark for each nibble<br><br>0110 0101 | | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(e) | Award **one** mark from:<br><br>• 01000 (1)<br>• 1000 (1) | | 1 |

## Type MS · Logical and arithmetic binary shifts

### Answer 2 of 5 · Question 1 part e · 1 mark

2021 JUN · Paper 01 · extract 2 of 2 · find part e in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(f) | 7 | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 1(g)(i) | Award **one** mark from:<br>• The smallest block (of colour) in an image (1)<br>• A single dot in a picture (1)<br>• A picture element (1)<br>• A single element of a digital screen (1) | | 1 |
| 1(g)(ii) | Award **one** mark from:<br>• 8 x 9 (1)<br>• 9 x 8 (1) | Accept 72 | 1 |
| 1(g)(iii) | 4 | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|

## Type MS · Logical and arithmetic binary shifts

### Answer 3 of 5 · Question 2 part c.i · 1 mark

2023 JUN · Paper 01 · extract 1 of 5 · find part c.i in this mark scheme extract

<table>
<tr>
<td>2(a)</td>
<td>Award one mark for each correct cell:
<table>
<tr><th>Denary</th><th>Hexadecimal</th></tr>
<tr><td>8</td><td>8 (1)</td></tr>
<tr><td>12</td><td>C (1)</td></tr>
</table>
</td>
<td></td>
<td>2</td>
</tr>
</table>

## Type MS · Logical and arithmetic binary shifts

### Answer 3 of 5 · Question 2 part c.i · 1 mark

2023 JUN · Paper 01 · extract 2 of 5 · find part c.i in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(b) | **The only correct answer is D**<br><br>***A*** *is not correct because* $6^2$ *stores 36 patterns, not 64 patterns*<br><br>***B*** *is not correct because* **6 × 2** stores 12 patterns, not 64 patterns<br><br>***C*** *is not correct because* $6^2$ **– 1** stores 35 patterns, not 64 patterns | | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(c)(i) | Award **one** mark for:<br><br>• 0001 0111 (1) | Ignore spaces | 1 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(c)(ii) | Award **one** mark for:<br><br>• 1111 1000 (1) | Ignore spaces | 1 |

## Type MS · Logical and arithmetic binary shifts

### Answer 3 of 5 · Question 2 part c.i · 1 mark

2023 JUN · Paper 01 · extract 3 of 5 · find part c.i in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(d)(i) | Award **one** mark for each correct nibble:<br>• 0011 1100 | Ignore spaces | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(d)(ii) | Award **one** mark for each of:<br>• Negative sign (1)<br>• 22 (1)<br><br>Example:<br>• -22 | Note 22 is calculated from 16 + 4 + 2 | 2 |

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(d)(iii) | Award **one** mark for each correct nibble:<br>• 0001 1011 | Ignore spaces | 2 |

## Type MS · Logical and arithmetic binary shifts

### Answer 3 of 5 · Question 2 part c.i · 1 mark

2023 JUN · Paper 01 · extract 4 of 5 · find part c.i in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|
| 2(e) | Award **one** mark for each of:<br><br>1. 13 x 1000 // 13000 in the numerator (1)<br>2. 1024 in the denominator (1)<br><br>Examples:<br>$\frac{13 \times 1000}{1024}$<br><br>$\frac{13000}{1024}$<br><br>$\frac{13 \times 500}{512}$<br><br>$\frac{13 \times 250}{256}$<br><br>$\frac{13 \times 125}{128}$<br><br>$\frac{13 \times 10^3}{2^{10}}$ | Award reduced expressions, if correct<br><br>Allow all equivalent exponents<br><br>Note 1000/1024 = 1.024, so could have 13 * 1.024 | 2 |

## Type MS · Logical and arithmetic binary shifts

### Answer 3 of 5 · Question 2 part c.i · 1 mark

2023 JUN · Paper 01 · extract 5 of 5 · find part c.i in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|

## Type MS · Logical and arithmetic binary shifts

### Answer 4 of 5 · Question 3 part b.ii · 1 mark

2024 JUN · Paper 01 · extract 1 of 4 · find part b.ii in this mark scheme extract

| | | | |
|---|---|---|---|
| 3(a)(i) | Award **one** mark for any of the following:<br><br>• 256 (1)<br>• $2^8$ (1)<br>• 2x2x2x2x2x2x2x2 (1) | | 1 |
| 3(a)(ii) | Award **one** mark for each of the following up to a maximum of **three** marks:<br><br>• 1920 x 1080 in the numerator (1)<br>• x 16/8 (16 in the numerator and 8 in the denominator) or (2 in the numerator) (1)<br>• / (1024 x 1024) in the denominator (1)<br><br>1920 pixels x 1080 pixels x 8 bits / 8 bits per byte / 1024 x 1024 bytes per MiB<br><br>$\frac{1920 \times 1080 \times \frac{16}{8}}{1024 \times 1024}$ | Max 2 marks if not a fully correct expression.<br><br>Allow equivalence e.g.<br>$1024 = 2^{10}$<br>$1024^2 = 2^{20}$<br><br>$2^{-20}$ is equivalent to $1/2^{20}$ | 3 |
| 3(a)(iii) | **The only correct answer is C**<br><br>***A** is not correct because the MSB is 1 so the denary value must be more than 128*<br>***B** is not correct because the LSB is 0 so the denary value must be even*<br>***D** is not correct because the denary number is signed* | | 1 |

## Type MS · Logical and arithmetic binary shifts

### Answer 4 of 5 · Question 3 part b.ii · 1 mark

2024 JUN · Paper 01 · extract 2 of 4 · find part b.ii in this mark scheme extract

| | | | |
|---|---|---|---|
| 3(b)(i) | Award **one** mark for each of:<br><br>• Negative sign (1)<br>• 63 (1)<br>Example:<br>- 63 | | 2 |
| 3(b)(ii) | Award **one** mark for:<br><br>1000 0010 (1) | | 1 |
| 3(b)(iii) | Award up to **two** marks for a linked explanation such as:<br><br>• An overflow error has occurred (1) because the result requires more bits to store than are available (1) | | 2 |
| 3(c)(i) | Award **one** mark for any of the following:<br><br>• 011 0101 (1)<br>• 53 (1)<br>• Hex value of 35 (1)<br><br>Do not accept 11 0101 (six bits, ASCII is at least seven) | | 1 |

## Type MS · Logical and arithmetic binary shifts

### Answer 4 of 5 · Question 3 part b.ii · 1 mark

2024 JUN · Paper 01 · extract 3 of 4 · find part b.ii in this mark scheme extract

| | | | |
|---|---|---|---|
| 3(c)(ii) | Award **one** mark for any of the following up to a maximum of **two** marks:<br>• It is a standard/universal encoding method (1)<br>• It can represent all the characters in the English alphabet (1)<br>• It only uses 7/8 bits **per character** (1)<br>• It can represent 128/256 characters (in standard/extended format) (1)<br>• It takes up less storage space than Unicode (1)<br>• It can be used with bit masking (1)<br>• It is compatible with UTF-8 (1) | | 2 |
| | | **Total for question 3** | **13** |

## Type MS · Logical and arithmetic binary shifts

### Answer 4 of 5 · Question 3 part b.ii · 1 mark

2024 JUN · Paper 01 · extract 4 of 4 · find part b.ii in this mark scheme extract

| Question Number | Answer | Additional Guidance | Mark |
|---|---|---|---|

## Type MS · Logical and arithmetic binary shifts

### Answer 5 of 5 · Question 3 part a.iii · 2 marks

2025 JUN · Paper 01 · extract 1 of 3 · find part a.iii in this mark scheme extract

| | | | |
|---|---|---|---|
| 3(a)(i) | Award **one** mark for each correct hexadecimal character in the correct order:<br><br>• B9 | | 2 |
| Question | Answer | Additional Guidance | Mark |
| 3(a)(ii) | **The only correct answer is B**<br><br>*A is not correct because both are processed at the same speed by a computer*<br>*C is not correct because both use the same amount of storage*<br>*D is not correct because hexadecimal uses number base 16* | | 1 |
| Question | Answer | Additional Guidance | Mark |
| 3(a)(iii) | Award **two** marks for a linked description such as:<br><br>• (logical/arithmetic) right shift (1) by 2 places (1) | | 2 |
| Question | Answer | Additional Guidance | Mark |
| 3(b)(i) | Award **one** mark for:<br><br>Pixel/picture element (1) | Accept additional words.<br>Accept incorrect spelling if word is recognisable as pixel | 1 |

## Type MS · Logical and arithmetic binary shifts

### Answer 5 of 5 · Question 3 part a.iii · 2 marks

2025 JUN · Paper 01 · extract 2 of 3 · find part a.iii in this mark scheme extract

<table>
<tr><th>Question</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>3(b)(ii)</td><td>Award one mark for each of:<br><br>• 16x16 pixels (1)<br>• Multiplying (16x16) by 2 bits per pixel in the numerator for colour depth (1)<br>• Dividing by 8 in the denominator to convert to bytes (1)<br>• Dividing 1000 by the number of bytes used for one image (1)<br><br>Example:<br><br>$1000 / \frac{16 \times 16 \times 2}{8}$ bytes<br><br>Allow follow-through</td><td>4 marks for any of:<br>1000 X 8 / 16 x 16 x 2<br>8000 / 256 x 2<br>1000 / 16 x 2 x 2<br>1000 / 16 x 4<br>8000 / 512</td><td>4</td></tr>
<tr><th>Question</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>3(b)(iii)</td><td>Award up to two marks for a linked explanation such as:<br><br>• high compression ratio (1) repeated/continuous runs (of the same pixel colour) (1)<br>• requires less storage space (1) encoding 16W for line 1 (1)</td><td>Allow valid RLE examples from the image.<br><br>Explanation can be a combination of different marking points. E.g. MP1a and MP2b</td><td>2</td></tr>
<tr><th>Question</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>3(c)</td><td>Award one mark for each correct cell:<br><br><table><tr><th>Encrypted text</th><th>Plain text</th></tr><tr><td>  </td><td>RED</td></tr><tr><td></td><td>COT</td></tr></table></td><td>Ignore capitalisation<br>Must be correct spelling</td><td>2</td></tr>
<tr><td colspan="3">Total for question 3</td><td>14</td></tr>
</table>



Match each answer number to the type practice PDF. Original exam material © Pearson Education Ltd.

## Type MS · Logical and arithmetic binary shifts

### Answer 5 of 5 · Question 3 part a.iii · 2 marks

2025 JUN · Paper 01 · extract 3 of 3 · find part a.iii in this mark scheme extract

| Question | Answer | Additional Guidance | Mark |
|---|---|---|---|